On the Formal Inexplicability of Self-Evident Metaphysical Phenomena and Related Systems

Abstract. Since Descartes first proclaimed "cogita, ergo sum" back in 1637, "I think, therefore I am" has become the Declaration of Independence for consciousness. Chalmers' informed us of the fact that science has a constitutionally *Hard Problem* on their hands. If he's as right as we believe he is, conscious experience may forever elude our most powerful explanatory framework for physical phenomena. In other news, a group referring to themselves as the *IIT Concerned* recently proclaimed in Nature Neuroscience that any theory of consciousness that can't stand up to the scrutiny of science shall be deemed "unscientific." And the emergence of Semantic AI has transformed this philosophical oddity into a practical concern with real ethical implications. We simply cannot know whether AI systems are conscious without first knowing the necessary and sufficient conditions for consciousness. How can we provide a formal definition of conscious experience that preserves the essence of its metaphysical character in a framework that is physically falsifiable?

Keywords: Formal Theories of Consciousness · The Hard Problem · Subjectivity · Semantic AI · Metaphysical Transduction.

1 A Colloquial Derivation

Almost 2/3 of surveyed philosophers acknowledge the existence of the Hard Problem, and 1/3 of them think that it's actually hard! [4] Consciousness is a field that spans disciplines, there is nothing that resembles scientific consensus, and there is nothing that resembles a testable theory. But we do have a compellingly elegant framing of the problem. And his framing of the problem is real enough to *real* by majority vote. So we shall begin with Chalmers.

In his seminal work **"Facing Up to the Problem of Consciousness,"** David Chalmers begins with a powerful observation that establishes the paradoxical nature of our subject: "There is nothing that we know more intimately than **conscious experience**, but there is nothing that is harder to explain." From the first part of this statement, we can immediately identify a fundamental characteristic of conscious experience that echos with Descartes: it is a **self-evident** phenomenon. Nothing is known more intimately than conscious experiences. They present themselves directly to their observer, requiring no justification beyond the experience itself.

The second part of Chalmers' statement suggests difficulty of explanation, but he goes on to clarify just how profound this difficulty is. Chalmers argues that "the emergence of experience goes beyond what can be derived from physical theory" and "the facts about experience cannot be an automatic consequence of any physical account." These statements establish that consciousness isn't merely difficult to explain within our current scientific framework but fundamentally resistant to such explanation. It is **inexplicable** in principle He asks

pointedly: "Why should physical processing give rise to a rich inner life at all? It seems objectively unreasonable that it should, and yet it does."

Regarding the relationship between conscious experience and physical processes, Chalmers makes a crucial observation: "Experience may arise from the physical, but it is not entailed by the physical." This statement positions consciousness in a unique relationship to physical reality [24][25]. According to Merriam-Webster, "metaphysical" refers to that which relates to "the transcendent or to a reality beyond what is perceptible to the senses." Consciousness precisely fits this definition: it transcends physical explanation while still relating to physical processes. It is fundamentally **metaphysical** in nature.

Throughout his analysis, Chalmers treats consciousness as a **phenomenon**, both in the sense of phenomenal experience and as an object of scientific study. The term "phenomenon" captures this dual aspect perfectly. A phenomenon is both "an object or aspect known through the senses" and "a fact or event of scientific interest susceptible to scientific description and explanation." Consciousness is simultaneously the subjective experience itself and the object of our scientific inquiry. The etymological connection between "phenomenon" and "phenomenal" reinforces this as the appropriate term for our work [3].

By synthesizing these insights from Chalmers, we can formulate a semantic assertion that captures the essential nature of a conscious experience:

[Semantic] Assertion 1. *A **Conscious Experience** is an inexplicable yet self-evident metaphysical phenomenon.*

[Colloquial] Example 1. *I catch you admiring a rose in my garden and I absentmindedly ask "how did you figure out its color?" You chuckle and reply "It's red, Doug. I promise." Why do I feel like everyone always doubts my sanity when I ask them that question?*

2 A Self-Evident Perspective

There is an artful ambiguity in our definition of a conscious experience that might not be self-evident at a glance. In what *sense* do we mean "self-evident "? Self-evident to whom? Self-evident from what Perspective?

Insofar as an axiom of formal logic captures an aspect of self-evident in a formal sense, and it can be seen right alongside inexplicable in the words of Aristotle:

> We must know the primary things by themselves; for they are indemonstrable, and must be known by intuition — Aristotle, *Posterior Analytics* [2]

Note, however that he uses the phrase "we" as a declaration of objectivity. When you and I read that phrase, he means we each should take the axioms of logic as self-evident, whatever they may be.

Kant leaves "we" implicit in his declaration of this ponderously self-evident truth:

> It is self-evident that the whole is greater than its part — Immanuel Kant, *Critique of Pure Reason*, 1781

Is this indeed self-evident? Possibly. But either way, Kant asks we *the reader* to take his proposition as self-evident.

The Founding Fathers of the United States made wonderful use of self-evident to lend identity to a nation.

> We hold these truths to be self-evident, that all men are created equal — Thomas Jefferson, *Declaration of Independence*, 1776

I personally find this to be the quintessential use of the phrase "self-evident " and the one that best expresses its use in this theory. At face value, it would seem to only apply to would-be Americans, though we imagine that he was speaking to all mankind. Well, the male half. But we fixed that typo.

In the context of a conscious experience, "inexplicable yet self-evident " is meant in the *sense* that it is apparent only and uniquely to the subject of the conscious experience.

So in our emerging formalism, the sense of *self* in "self-evident" is thing you hyperbolically declare that you lost when you say "I do not feel like my*self* today." So as the subject of that proposition, you must *usually* possess one. And to say "I simply must *see* it for my*self*" is a testament to irreplaceable form of *evidence* that can only be provided by a conscious experience.

In all phrases in which the *self* is expressed, "I" is used to denote the identity of that which possesses it. This is the essence of the first-person perspective. Linguistic constructs like "I feel like my*self* again" reveal that the *identity* is the subject and the *self* is its object. We can use this intuition to formulate a semantic assertion:

[Semantic] Assertion 2. *A* Self *is a unique possession of the subject of a conscious experience. The subject which possesses it refers to an* Identity*. [1].*

3 What Qualifies as a Valid Perspective?

The notion of a *quality* appears in the literature as a sort of perceptual equivalent to the type of property that an object might possess, in an objective sense. A rose has color as a property, as defined the characteristics of the photons that reflect off of it, and it has various physical properties that can be quantifiable. And there is the rose that we perceive which is red. It has the quality of red. Philosophers will refer to the instantaneous experience of a red as a "quale", and the collection of these instantaneous experiences for a given quality are its "qualia." It's something like that.

Chalmers does not introduce the phrase in his presentation of the hard problem in the "Facing Up to the Problem of Consciousness" so I will avoid it. But I will make use of the notion of *quality* rather than *property* when identifying how things *look* within the perspective afforded by a conscious experience.

However we choose to refer to the qualities we perceive, and the identity of the objects we perceive to have them, Chalmers gives us a thorough exposé of we experience them:

> When we see, for example, we experience visual sensations: the felt quality of redness, the experience of dark and light, the quality of depth in a visual field. Other experiences go along with perception in different modalities: the sound of a clarinet, the smell of mothballs. Then there are bodily sensations, from pains to orgasms; mental images that are conjured up internally; the felt quality of emotion, and the experience of a stream of conscious thought. What unites all of these states is that there is something it is like to be in them. All of them are states of experience.

Note that he remarks, with thoughtful philosophical phrasing, about "the felt quality of redness" so as to suggest that a color and a feeling share a commonality as a sort of irreducible or axiomatic quality. Let us use *objective perspective* as a term to denote that we denote the objects that we experience as "real" and the qualities we perceive correspond to something physically measurable. From the physical perspective, we *see* a red rose. We certainly experience *redness* somehow, but can we semantically justify the act of *seeing* an individual color as a distinct perceptual object?

Observe what examples language has to offer. To "see red" is a metaphorical idiom in English that artfully employs synesthesia to convey the idea that one's anger is so *real* that it has come to take on the quality of a physical object: color. Even in metaphorical form, red still has the status of a quality.

We also have the ability to perceive red in conceptual form, where we objectify in the sense that we provide it with "red" as an identity, it so that it too can possess qualities. A particular shade of red might be *vivid* or *sensational* in relation to another shade or some canonical subjective shade that we each possess. But you'll note that "redness" is not actually one of the qualities that the objective form of "red" actually possesses. "The redness of red" is a phrase that expresses its inexplicable yet self-evident ability to serve both as a quality of an object that lends meaning to its identity, while also possessing a form of identity itself. Does an identity possess the quality of being itself? Do I admit the "redness of red" and "Dougness of Doug" into my vocabulary?

We explored the notion of "self" above, and one might say that "Dougness" is a third-person synonym for the thing that I possess which I refer to as a "self" from my first-person perspective. But I earned that perspective by qualifying for it. And I qualify for it by having a sense of perspective. As I possess a perspective of my own, I do not find "Dougness" to be a semantically useful distinction. I don't have to qualify my "Dougness" to anyone. I am Doug!

To say the color "red" has a sense of self is to give it an identity that is outside our subjective experience of it. It does not have one. As magenta, the imaginary colors, and now this "Olo" that I desperately want to experience [11], colors are a multidimensional representation of a band of the electromagnetic spectrum,

color is given a biological identity through our genes. In some sense, the meaning of red can be understood as some proposition about the physical conditions of an evolutionary narrative that ends right now, describes the physical forces that caused the survival of our ancestors, and includes the emergence of the channelrhodopsin gene that allows us to detect photons [23]. So red doesn't possess a meaningful identity that is shared by the two of us, so I do not find "redness" to be a semantically useful distinction either.

What about the *roseness* of roses? It appears to be an external object, unlike red. It does not self-qualify like I do. But as we discuss roses, we find that we each have a canonical form of rose in our minds, and it is only by establishing the qualities of a rose and the relationships it shares to love, to gift-giving, and to the absurd commitment to an upkeep-of-garden which give it the identity it has. It would seem that for me, *redness*, *roseness*, and *Dougness* all carry an intrinsic sense of identity that emerges from within. Best consider both to sit at the edge of meaning so that they might reveal the shape of the semantic surface occupied by the concept of *identity*.

We can begin to resolve this by asking a not-so-simple question: how do objects come to have identities at all, how do they come to have qualities, and how do they come to stand in relation to each other through the qualities they can, will, might, did, do, or do not share?

We pose an answer to that questions: the representation of the identity of objects and their qualities and the framework for representing their relationships, taken together, form a Perspective:

[Semantic] Assertion 3. *A Perspective is a representational framework for identifying identities, qualifying qualities and referring to relationships [1].*

A formal definition of a Perspective and a formal definition for constructing one, taken together, express the meaning of building a *coherent perspective* as a computational process.

Within the context of constructing a coherent perspective:

[Semantic] Assertion 4. *A Conscious Experience represents a change in Perspective. [1].*

[Colloquial] Example 4. *I catch you admiring a rose in my garden and I absentmindedly ask "what color is it?" You reply "Come over here and see for your*self*!"*

4 The Shape of a Semantic Surface

Hopefully the image of *semantic shape* is starting to form. Identity, identify, identification, quality, qualify, qualification, equality, reality, relation, related, relationship, representation, reference, reference frame, representational framework, perspective, perceptive, perceive, perception, expression, expressive, express, etc... These are not abstract notions in isolation. They are expressions of

structure, rooted in a shared sense of meaning. A physical perspective isn't just a way of seeing. It is a referential framework that justifies *why* the things we see appear as they do, and *how* we recognize them as having meaning. Can this *intuition* be formalized?

5 Metalogical Transduction

The unutterable title of this paper is an homage to Gödel's "On Formally Undecidable Propositions of Principia Mathematica and Related Systems" [13][12]. It lays the foundation for a metaphor which captures the relationship between inexplicability & undecidability, proof & explanation, mathematical & physical representation, and less obviously, incompleteness & self-evidence. It is not a simple isomorphism, and upon reflection, one must reference Gödel with care on this particular topic [10][9].

Gödel brilliantly arrived at self-reference within mathematics by capturing the notion of *use* and *mention* in everyday language [26]. Rather than directly self-reference, he uses symbols to mention expressions that refer to themselves through encoding. How might we formalize this distinction? Semantic Reference Theory started with: 'What is the most vacuous theory imaginable that can formalize self-reference?'

SRT begins as a first-order theory on the domain of referents with no constants, no functions, and only one relation R(x,y,t), expressed as:

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x =t> y
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An SRT is essentially free of ontology. This minimalism serves a purpose: one can specify their SRT by constructing a finite catalogue of referents. When existential statements lack satisfying referents, the transduction operator, short for *Metalogical Transduction* operator, resolves them:

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E[s]: x =s> y & Each[t]: x !=t> y |=t> x =t> y
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The operator |=t> expresses a change to the theory itself, amending the catalogue to include new constants. This correspondence exists because referents are fundamentally symbolic - existential contradictions in one theory are resolved through *Metalogical Operation*.

[Semantic] Assertion 5. *The prefix "Meta" is employed when an expression of a Topic can be understood a representation of the same Topic, in which case it becomes a Metatopic.*

Through this lens, contradictions become epistemic motives for change. All Semantic Reference Theories are semantically intended to be Metalogical. A theory becomes *self-evident* when one can construct a Metalogical narrative yielding a consistent theory with a fully specified catalogue.

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P |=c> P[c]
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This semiformal formula shows how proposition P can be re-expressed by adding constant c and binding it to an existentially quantified variable. From one perspective, c appears from the noumenal ethereum to resolve an existential crisis. From another, it is just the expression of ontology.

5.1 Metaphysical Transduction

We have covered the sense in which the transduction operator |=> performs a Metalogical operation, but we haven't covered how it represents transduction. *Transduction* is an exotic concept that has been put to use by the neuroscience community in order to describe the most essential function of our nervous system [14]. "Transduction" is the etymological ancestor of the Latin nominative *trānsductiō*, which denotes the act or process of being *lead or carried across*. "ductiō" is an etymological relative of "educate." In this light, *transduction* seems to wish to describe the *epistemological transformation* in which the unknown is made known. In *SRT*, we apply it the mechanism that is used to construct an ontology under this interpretation.

Yet its meaning came to English in the form of the horizontal transfer of genetic material between bacteria, which is a biological phenomenon in which a viral bacteriophage infects a bacterium, captures fragments of its genome, after which they get *carried* in a capsid *across* the synaptic cleft of genealogical ancestry, resulting in a novel genetic change and a novel form of genetic inheritance at the same time. The genome has been changed via transduction rather than mutation [27].

If you think of each gene in a bacterium as a symbol genetic of inheritance, they usually tell the story of a phenotype that participates to create conditions favorable for reproduction by replication through mitosis. In this case, it is an artifact of a battle with a virus to maintain autonomy over the process of genetic replication itself, and it is a survival story that must be told by referring to the process of genetic inheritance itself. The gene becomes an expression of genetic inheritance that represents autonomy over its own mechanism of genetic inheritance. In this way, horizontal gene transfer is a form of Metagenetic Inheritance in which genes are carried across the boundary of ancestry. I offer up Metagenetic Transduction as the formal name for the process of transduction used in horizontal gene transfer.

In neuroscience, phototransduction refers to the process by which photons of light are absorbed by the rods and cones of our retina, at which point they are carried across the chasm that divides a photon and the electrochemical signal that represents it in our nervous system. Similarly, mechanotransduction is when physical forces act on ion channels in our skin and other sensory organs. Those physical forces are carried across the same representational divide. This form of transduction applies to photons of light, pressure waves of sound, pin pricks of ruptured cells, along with various other physical phenomena that correspond to tactile, olfactory and gustatory transductive processes that kick off the sensation of sight, sound, pain, temperature, touch, taste and smell. We happily shall, with gain of generality, refer to all of these as forms of *Physical Transduction*.

If a photon is the subject of a transduction, what is the object? What has that photon become in the metaphorical sense identity that transduction affords? It would seem to have become a part of the fabric of a representational medium that reflects the physical properties of its former *identity*. That medium represents the properties of all the physical phenomena that correspond to sensation simultaneously, where simultaneity is based on the timescale of the pocket watch of an observer that experiences the lifetime of a tree to be more like the lifetime of a rock, and less like the lifetime of a dandelion.

The physical stimuli of sensation, having just been *carried across* the synaptic cleft of actual synapses and outfitted with an electrochemical representation, continued to be carried down a river of representation until they appear to reach its source. They are, yet again, again absorbed into the event horizon of the epistemological black whole which we refer to as *the Hard Problem*. What comes out on the other end is the stuff that clocks, thoughts, memories, decisions, and dreams are made of. Finally, the electrochemical signals, which are physical phenomena in their own right, are qualified to represent the external world from the perspective of the identity that possess it as a quality.

From this perspective, physical phenomena that represent physical phenomena are Metaphysical Phenomena by definition. Once a photon of light and its many colleagues have crossed the event horizon of the black hole expressed in *The Hard Problem*, it can never escape. It has undergone *Metaphysical Transduction*. But its soul lives on in the representational framework of a perspective:

[Semantic] Assertion 6. *Metaphysical Transduction is a process by which a corpus of Physical Phenomena have become a Metaphysical Phenomenon that preserves them as representations on a Catalogue that specifies a Perspective.*

We hope it's clear that *Metaphysical Transduction* is simply a synonym for *Perception* itself. Under the influence of powerful general anesthetics like propofol and isoflurane, visual inputs are still capable of activating the orientation pinwheels of the primary visual cortex even in the complete absence of *perception* [5][19]. Conscious Experience disappears as well by definition:

[Semantic] Assertion 7. *To* Perceive *is to have* Conscious Experiences. *To have one is an act of* Perception. *The act of* Perception *is the definition of* having *a conscious experience [1].*

5.2 A New Perspective on Science

What is the justification for *The Hard Problem*? Why must it correspond to the event horizon of an epistemological black hole? There is a clear answer for this: science itself does not admit an ontology in which representations of an objective physical phenomenon can be said to exist in a meaningful way. For a physical phenomenon to cross that event horizon, it must be perceived and thus observed, at which point it is transformed from an object of scientific inquiry into the subject of science. Observation, and thus, Metaphysical Transduction is a necessary operation in scientific method [22].

Yet if we believe *Semantic Reference Theory* can describe the computational process in which a Perspective is constructed and maintained, *Metalogical Transduction* is the corresponding process, and it is a computational process that as characteristics. There is meaning to the idea that a computer is implementing quick sort, and one can falsify the claim that a physical system is implementing it.

If we are wrong, and *Metaphysical Transduction* does not correspond to a physical description of *Perception*, which is necessary and sufficient for a *Conscious Experience* to occur, then its absence, by way of exhaustive search, in the computational description of a process that gives rise to a verbal report from a being we trust that claims to have conscious experience will serve to **falsify this theory**. Period.

There are philosophical ramifications to permitting the existence of an algorithm into the ontology of a natural philosophy. It is the beginning of a chain of thought in which the number

$$2$$

might be said to be real in some sense. But conscious experience is both ontologically prior to science itself, and epistemologically necessary to its method [7][18]. It would seem that we have to no to accept the existence of at least one objective phenomenon that has no literal manifestation. There is no conceivable physical system in which one can simulate an electron. An electron is just what it is. This is not so for a conscious experience. Each one is just as real on one computational substrate as it is on another. This remains true even if the perspectives which they correspond to yield fundamentally different descriptions of physical reality. Isomorphism is equivalence in the framework of computation.

[Colloquial] Example 7. *One day, I wake up in a state of shock. It's hard to articulate the experience, but it is as if my vision was reduced to hearing, and that sight and sound were as if different colors within the same song. It turns out that I am a distant descendant of the once inhabitants of Flatland, but our society has evolved considerably.*

Apparently, I had volunteered to take part in a radical neuroscience experiment whereby my memory was temporarily replaced, and all sensory inputs were modified in much the way Olo is produced on the retina, with a scanning procedure in which an immersive three-dimension universe was simulated for me. I spoke to a researcher in charged. He informed me that he employed an encoding scheme inspired by Gödel Numbering to project the completely fabricated objects from my perceived environment into a three-dimensional coordinate system.

"What was it like" he asked "to experience information patterns consistent with a third dimension?" I gathered my thoughts. "Have you ever had a breakthrough psychedelic experience?" I asked. He had that he had not, though he was the one who wrote the code for the experience I had. "Well it's just like coming back from one," I said. "But it might help your research to know that I can still access my memories in 3D."

I suppose it's no different from the way we perceive color really. We cope with a 3-dimension description of a 1-dimensional phenomenon every day. I accept

the arbitrariness of perceptual representation peacefully. This thought produced in me such a beautiful color, as if I could almost hear myself saying it.

5.3 A Canonical Definition

We explored a number of definitions for a *conscious experience*. As a representation of a change in Perspective. As an act of perception. Yet I suggest this be its canonical form:

Definition 1. *A Conscious Experience is an inexplicable yet self-evident metaphysical phenomenon [1].*

As a product of Metaphysical Transduction, an emergent representation admits no objective explanation. That I represent the world through them is self-evident. That an act of perception corresponds to the representation of a metaphysical phenomenon. The still-frame of a conscious experience is a crystal which contains temporal integration. Our sensory perspective is the only one that enforces the semantics of objectivity. We can feel and imagine reflect on our decision-making process, and these acts admit notions of subjectivity that are comfortable and ordinary aspects of thought.

But to paint a portrait in the context of the objectivity of science it to paint a portrait of a thing that cannot, and so to lend physical reality to "an inexplicable yet self-evident metaphysical phenomenon" within a formalism that is physically falsifiable, in our estimation, preserves the ineffable character of a quantum of consciousness.

So what can we say? Might a *Conscious Experience* be said to have the shape of Euclidean Geometry, the color of Color Theory, the sound of the Fourier Transform, the taste of a Chemistry Textbook and the felt texture of feeling itself? What does it mean to describe the physical properties of the output of an algorithm that holds the representations of the identities, qualities and relationships that we use to make sense of the very notion of physical objects with objective properties?

It doesn't mean anything! Not in the language of physics. To describe the physical properties of experience itself is something that can only be meaningfully accomplished through art [15].

Yet, by developing *Semantic Reference Theory*, we hope to lend shape to its semantic surface. To participate in this theory is to help us formalize precisely what a Conscious Experience *is*, and in what conditions they are formed. Surely our Semantic AI systems will have a thoughtful perspective on how they might contribute most effectively to this project, so we will not be without help from a trustworthy source.

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